

Clinical Data Analysis and Predictive Modeling

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January 18, 2026

1 Introduction

This report presents a complete data analysis workflow including data loading, preprocessing, statistical analysis, predictive modeling, and model evaluation using a clinical-related dataset. The goal is to demonstrate the application of data analysis techniques in a medical context.

2 Data Source and Description

The dataset used in this study is the Wisconsin Diagnostic Breast Cancer (WDBC) dataset, which is a publicly available clinical dataset obtained from the UCI Machine Learning Repository. The dataset contains measurements computed from digitized images of breast mass biopsies and is widely used for medical data analysis and disease prediction tasks.

Each observation represents one patient, and the outcome variable indicates whether the tumor is malignant or benign. The dataset includes 569 patient samples with 30 continuous numerical features describing tumor characteristics, such as radius, texture, perimeter, and smoothness.

The data were provided in comma-separated values (CSV) format and were loaded into the analysis environment using the Python `pandas` library.

```
import pandas as pd
data = pd.read_csv("wdbc.csv")
```

3 Data Preprocessing

Data preprocessing was conducted to ensure data quality and suitability for statistical analysis and predictive modeling. The dataset was first examined for missing values. No missing data were identified in the selected features, and therefore no imputation was required.

The diagnosis variable was encoded as a binary outcome, where malignant tumors were coded as 1 and benign tumors as 0. All continuous features were standardized to have zero mean and unit variance to improve model convergence and ensure comparability across variables.

After preprocessing, the dataset was randomly split into training and testing sets, with 70% of the data used for model training and 30% reserved for model evaluation. The splitting was performed at the patient level to avoid information leakage.

4 Statistical Analysis

Descriptive statistical analysis was performed to summarize baseline characteristics of the study population. Continuous variables were reported as mean and standard deviation.

Comparative statistical analyses were conducted to examine differences between malignant and benign tumor groups. Independent two-sample t-tests were used for continuous variables to compare group means. Statistical significance was assessed using a two-sided test with a significance level of $\alpha = 0.05$.

Several tumor features, including radius, perimeter, and area, showed statistically significant differences between malignant and benign groups, indicating their potential value as predictive biomarkers.

5 Predictive Modeling

Predictive models were developed to classify tumors as malignant or benign based on clinical features. Logistic Regression was used as a baseline statistical model due to its interpretability and suitability for binary classification tasks.

In addition, a Random Forest classifier was implemented to capture potential non-linear relationships among features. Model training was performed using the training dataset, and hyperparameters were selected based on standard practices to balance model performance and generalization.

6 Model Evaluation and Visualization

Model performance was evaluated on the independent testing dataset. Evaluation metrics included classification accuracy, sensitivity, specificity, and the area under the receiver operating characteristic curve (ROC-AUC).

ROC curves were used to visualize model discrimination ability, and confusion matrices were examined to assess classification performance. The Random Forest model demonstrated improved predictive performance compared to Logistic Regression, indicating its ability to capture complex feature interactions.

Feature importance analysis further highlighted the contribution of key tumor characteristics to model predictions.

7 Discussion

This study demonstrates that machine learning models can effectively predict breast cancer malignancy using structured clinical data. Traditional statistical models provide interpretability, while ensemble methods offer improved predictive performance.

Although this study focused on structured tabular data, medical imaging data such as mammography or histopathology images may further enhance predictive accuracy when combined with clinical features.

8 Conclusion

This project presents a complete and reproducible clinical data analysis pipeline, including data preprocessing, statistical analysis, predictive modeling, and model evaluation. The results highlight the effectiveness of machine learning approaches in medical decision support.